

Syllabus : Numerical Analysis (MAT 211)

Chapter 1: Error Analysis

Identifying Error Type, Error calculation, Error propagation for single variable and multiple variable formula. (Follow class works, home works and class test problems)

Chapter 2: Root Finding

Bisection Method, False Position/Regula – Falsi Method, Iteration Method, Newton Raphson Method. (Lecture note problems)

Chapter 3: Interpolating Polynomials

Equal Interval - Newton's Forward Difference formula, Gauss's Forward Interpolation formula.

Unequal Interval – Lagrange Interpolation, Newton's Divided Difference.

Problems: Related Solved Example only in the referred textbook.

Chapter 4: Numerical Differentiation

Solved Examples: 9.1, 9.2, 9.3 (Using Newton's Forward and Backward)

Chapter 5: Numerical Integration

Methods Studied at class (Except Weddle's)- Related Solved Examples only in the referred textbook.

Chapter 6: Solving System of Linear Equation:

Gauss Jordan, Factorization, Gauss Jacobi/Gauss Seidel (Related Solved Examples from the book)

Chapter 7: Solving ODE

Euler, Improved Euler, Modified Euler, 4th order RK method.

For the first three methods: Examples: 11.9, 11.10, Exercise: 6,8

For RK4 method:

- [1] Solve $\frac{d^2y}{dx^2} = x - y$ with $y(0) = 1, y'(0) = 0$ and $h=0.5$ Using 4th order RKM.
- [2] Solve $\frac{dy}{dx} = x + y$ where $y(0) = 1$, find $y(0.2)$ using 4th order RKM.
- [3] Find $y(0.3)$ using 4th order RKM for the following differential equation $\frac{dy}{dx} = -y - xy^2$, where $y(0) = 1$